

14.24 Solution: Comparing the form of Liénard-Wiechert potentials to the form of static results, we can see that there is a difference in the factor of

$$\frac{dt}{dt'} = 1 - \mathbf{n} \cdot \boldsymbol{\beta}.$$

This suggests that we should consider an integral along the loop in order to cancel this factor.

The circuit can be divided into N intervals of length Δ , so that there is only one particle in each interval at any instant. At time t , the scalar potential can be written as the total contribution from these intervals,

$$\Phi(\mathbf{x}, t) = \sum_{k=1}^N \frac{q}{(1 - \mathbf{n}_k(t_{\text{ret},k}) \cdot \boldsymbol{\beta}_k(t_{\text{ret},k})) R_k},$$

where the retarded time is $t_{\text{ret},k} = t - R_k/c$, with $\mathbf{x}$ and $\mathbf{x}'_k$ as the location of the observer and the k -th interval, respectively, and $R_k = |\mathbf{x} - \mathbf{x}'_k(t_{\text{ret},k})|$.

We now want to turn the above summation to an integral. To this end, we have the equivalent expression

$$\Phi(\mathbf{x}, t) = \frac{q}{\Delta} \sum_{k=1}^N \frac{\Delta}{(1 - \mathbf{n}_k(t_{\text{ret},k}) \cdot \boldsymbol{\beta}_k(t_{\text{ret},k})) R_k}.$$

In the limit of $N \rightarrow \infty$, every interval can be viewed as a straight line segment,

$$\Delta \approx v \delta t,$$

where δt is the time in the lab frame for a particle to traverse the k -th interval, and is related to the retarded time interval on the k -th interval by

$$\frac{\delta t}{\delta t_{\text{ret},k}} \approx 1 - \mathbf{n}_k(t_{\text{ret},k}) \cdot \boldsymbol{\beta}_k(t_{\text{ret},k}).$$

Therefore,

$$\Phi(\mathbf{x}, t) = \frac{q}{\Delta} \sum_{k=1}^N \frac{v \delta t_{\text{ret},k}}{R_k} \rightarrow \frac{q}{\Delta} \oint \frac{v dt_{\text{ret},k}}{|\mathbf{x} - \mathbf{x}'(t_{\text{ret},k})|},$$

in the $N \rightarrow \infty$ limit. Finally, we can relabel the time stamp for each interval with the non-retarded time, as there is a one-to-one correspondence. Define the linear charge density as $\rho = q/\Delta$ and the length element $dl = v dt$, we will have

$$\Phi(\mathbf{x}, t) = \frac{q}{\Delta} \oint \frac{v dt}{|\mathbf{x} - \mathbf{x}'(t)|} = \oint \frac{\rho dl}{|\mathbf{x} - \mathbf{x}'|}.$$

Therefore, for a steady current, the scalar potential is identical to the static result. We can obtain the same result for vector potential following the same argument. Then, for a steady current, there will be no radiation.